INCH-POUND

A-A-55271
22 March 1995

COMMERCIAL ITEM DESCRIPTION

CLOTH, WIND RESISTANT, POPLIN, COTTON

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-C-342 for all Federal agencies.

1. SCOPE. This commercial item description covers rip stop poplin, wind-resistant cotton cloth. The cloth is intended for use in the manufacture of clothing and equipage items. The class B cloth, although water- repellent treated, will not provide rain protection due to its high air permeability. The class C cloth is intended to be used as a component of the "CVC" face mask.

2. CLASSIFICATION. The cloth shall be of one type in the following classes:

Class A - Plain finish (untreated)
Class B - Water-repellent treated
Class C - Fire-resistant treated

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FQSD, 2800 South 20th Street, Philadelphia, PA 19145-5099.

AMSC N/A FSC 8305

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3. SALIENT CHARACTERISTICS

3.1 General description. The woven cloth shall be made of cotton which has been carded, combed, drawn, spun and twisted into 2-ply yarn for the warp and 2-ply or singles yarn for the filling. The weave shall be plain weave with reinforcement ribs uniformly spaced in the warp and filling. The warp repeat shall be two ends weaving as one followed by 23 single ends (total of 25 ends per repeat). The filling repeat shall be two picks inserted into the same shed followed by 12 single picks (total of 14 picks per repeat). The class A cloth shall weigh 5.7 to 7.0 ounces per square yard; the class B cloth shall weigh 5.7 to 7.2 ounces per square yard; and the class C cloth shall weigh 7.0 to 8.0 ounces per square yard when tested according to ASTM-D 3776, Option C. All three classes of cloth shall contain a minimum of 104 warp ends and 52 picks per inch when tested according to ASTM-D 3775. Prior to testing for yarns per inch, the samples shall be conditioned as directed in ASTM-D 1776 - Practice for Conditioning Textiles for Testing. The width of the cloth shall be as specified in the contract or purchase order, inclusive of the selvage when fly shuttle looms or shuttleless looms with tuck-in selvage looms are used. The color shall be as specified in the contract or purchase order and shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7500° ± 200° K, with illumination of 100 ± 20 foot candles and shall be a good match to the standard sample under incandescent lamplight at 2300° ± 200° K.

3.2 Physical requirements. The finished cloth shall conform to the following physical requirements.

3.2.1 The cloth shall be singed, desized, mercerized and dyed. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid and the use of pigments in the dyeing or finishing of the cloth are prohibited. When class B finish is specified, the cloth shall be given an approved Quarpel-type, water-repellent treatment. When class C finish is specified, the cloth shall be given an approved flame-resistant treatment.

3.2.2 The minimum breaking strength for the finished cloths shall be 110 pounds in the warp and 68 pounds in the filling for class A, 100 pounds in the warp and 60 pounds in the filling for class B, and 90 pounds in the warp and 60 pounds in the filling for class C when tested according to ASTM-D 5034. The minimum tearing strength for the finished cloth shall be 4.0 pounds in both the warp and filling for class A and B; and 2.0 pounds in the warp and 3.0 pounds in the filling for class C when tested according to ASTM-D 1424.

3.2.3 The maximum dimensional change for the cloths shall be 2.0 percent in both the warp and filling for all classes when tested according to AATCC 96, Option Vc, with the following exceptions: Sample size shall be 22 by 22 inches with 18 inch benchmarks. Use Drying Procedure (D), or samples may be dried using a thermostatically controlled circulating oven at 221° to 230° F until dried. The sample shall be allowed to cool for 5 minutes, sprinkled with water to permit

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damp pressing, and allowed to stand in this condition for 5 minutes. The dampened specimens shall be pressed with a flatbed press as described in AATCC 96, Section 5.2, or a hand iron as described in AATCC 96, Section 5.10, at a temperature of 275° to 300° F. Section 7.4 of AATCC 96 shall not apply.

3.2.4 The seam efficiency for classes A and B shall be 80 percent minimum when measured according to Method 5110 of FED-STD-191 - Textile Test Methods. The needle shall measure 0.040 inch (± 0.001 inch) across the blade at the eye. The cotton thread shall be type IA1 conforming to A-A-52094, ticket No. 40-3 in the needle and ticket No. 70-3 ply in the loopers.

3.2.5 The cloth shall measure an average maximum Air Permeability (cu.ft./minute/sq.ft.) of 18.0 for Class A and 15.0 for Classes B and C according to ASTM-D 737.

3.2.6 The maximum percent of non-fibrous material for all classes shall be 2.0 percent when tested according to ASTM-D 629 Section 9. The pH value for all classes shall be 5.0 to 8.5 when tested according to Method 2811 of FED-STD-191 - Textile Test Methods. The cloth shall contain no more labile sulfur than that shown by the standard sample. When no standard is available, the finished cloth shall show no more than a slight trace of labile sulfur.

3.2.7 For Class B finished cloth, the initial spray rating on a 7 inch by 7 inch sample, after three individual determinations, shall be equal to or better than 90,90,80 when tested according to AATCC 22. The spray rating after laundering, after three individual determinations, shall be 70,70,70 when tested according to AATCC 96, Option Vc with the following exceptions: Sample size shall be 22 by 22 inches with 18 inch benchmarks. Use Drying Procedure (D), or samples may be dried using a thermostatically controlled circulating oven at 221° to 230° F until dried. The sample shall be allowed to cool for 5 minutes, sprinkled with water to permit damp pressing, and allowed to stand in this condition for 5 minutes. The dampened specimens shall be pressed with a flatbed press as described in AATCC 96, Section 5.2, or a hand iron as described in AATCC 96, Section 5.10, at a temperature of 275° to 300° F. Section 7.4 of AATCC 96 shall not apply.

3.2.8 For Class C, the finished cloth shall have an average time for flaming of not more than 2.0 seconds, an after-glow of not more than 6.0 seconds, and an average char length of not more than 5.0 inches both initially and after 15 launderings when tested in accordance with Method 5903 (Flame Resistance of Cloth; Vertical) of FED-STD-191 - Textile Test Methods. The 15 launderings shall be performed in accordance with Method 5556 (Mobil Laundry Evaluation for Textile Materials) of FED-STD-191 with the following exception: the specimens must be dried after each laundering cycle.

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3.2.9 The finished cloth shall show colorfastness to laundering (after 3 cycles), perspiration, wet dry-cleaning, crocking and light equal to or better than the standard sample. As a limit of acceptability or when no standard sample is available, the finished cloth shall show a minimum rating of "4" or (good) colorfastness to light, laundering and perspiration (acid and alkaline) and a minimum rating of "4" (good) colorfastness to dry crocking and "3" (fair) colorfastness to wet crocking when tested in accordance with the indicated test methods:

AATCC 61, Test 3A - Colorfastness to Laundering with the following exceptions: On the color transfer cloth evaluation, only the stain on the cotton fibers of the color transfer cloth shall be evaluated. The soap solution shall contain 0.5 percent by weight, P-D-245 - Detergent, Laundry, and Hand Dishwashing (Granular), Type II, in water hardness not over 50 parts per million. The weight of the specimen shall be 4.0 $\pm$ 0.5 grams and the rotating time shall be 30 minutes with 10 stainless steel balls in each cylinder.

AATCC 15 - Colorfastness to Perspiration with the following exception: The test for colorfastness for acid and alkaline solution shall apply.

AATCC 132 - Colorfastness to Wet Dry-cleaning.

AATCC 8-1989 - Colorfastness to Crocking.

AATCC 16, Option A - Colorfastness to Light after 40 AFU's.

3.2.10 Spectral reflectance data shall be obtained from 600 to 860 nanometer (nm) at 20 nm intervals on a spectrophotometer relative to a barium sulfate standard, the preferred white reference standard. Other white reference materials may be used provided they are calibrated to absolute white; e.g., Halon, magnesium oxide or vitrolite tiles. The spectral bandwidth shall be less than 26 nm at 860 nm. Reflectance measurements may be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a source that simulates either CID Source A or CIE Source D65. The specimen shall be measured as a single layer, backed with four layers of the same fabric and shade. Measurements will be taken on a minimum of two different areas, and the data averaged. The specimen shall be viewed at an angle no greater than 10° from normal, with the specular component included. Photometric accuracy of the spectrophotometer shall be within 1 percent, and wave length accuracy with 2 nm. The standard aperture size used in the color measurement device shall be 1.0 to 1.25 inches in diameter. When the measured reflectance values for any color at four or more wavelengths do not meet the limits specified, it shall be a test failure. When Camouflage Green 483 is specified, the finished cloth shall meet the spectral reflectance factors (in percent) for the visible/near infrared wavelength range 600 to 860 nanometers (nm) as indicated in Table I.

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TABLE I. Spectral reflectance

Wavelength (nm)	Reflectance (%)		Wavelength(nm)	Reflectance (%)	
	Max.	Min.		Max.	Min.
600	10	4	740	40	14
620	10	4	760	49	18
640	11	4	780	55	23
660	13	4	800	60	29
680	15	4	820	64	34
700	20	6	840	67	39
720	30	9	860	69	45

3.3 Length and put-up. The coated cloth shall be furnished in rolls in continuous lengths as specified in FED-STD-803 - Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics), Packaging of, of not more than 175 nor less than 150 yards in length. No roll shall contain more than six pieces and no individual piece shall be less than six yards in length. The ends of the pieces shall be overlapped, not joined by a seam.

3.4 Label/tag. Each piece shall be labeled, or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act. The face side of the cloth shall be marked with the word "FACE" at each end of the piece. This label/tag shall be located so that it is completely visible on the roll when it is packaged as specified and so that it causes no damage to the roll and that it remains on the roll as cloth lengths are distributed.

4. REGULATORY REQUIREMENTS

4.1 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices, and is the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract or purchase order.

5.2 Market Acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this commercial item description.

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5.2.1 The company offering the cloth furnished in continuous lengths, or generically similar cloth as described herein, must have been sold to the commercial marekt or to the Government for at least 2 years.

5.3 Warranty. The item offered shall include the standard warranty given to the commercial market beginning with the date of delivery of the individual item.

5.4 Visual examination. The cloth shall be examined as indicated in 5.4.1 through 5.4.1.4 listed below:

5.4.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined as Section II of FED-STD-4 - Glossary of Fabric Imperfections, which are clearly visible at normal inspection distance (3 feet) shall be scored and assigned demerit points as specified in the contract or purchase order, except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale "D" or 3-1/2 for slubs and "C" for knots, shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. For determination of acceptability on the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards exceeds the point value as specified in the contract or purchase order. For determination of acceptability on the individual roll quality basis, the lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed the point value as specified in the contract or purchase order. If one roll exceeds the point value per 100 square yards, a second sample of 20 rolls shall be examined for individual roll quality examination. If one or more rolls in the second sample exceeds the point value, the lot shall be rejected. Point computation for the lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yds. inspected}} = \text{Points per 100 sq.yds.}$$

5.4.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point.
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points.
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points.
For defects exceeding 9 inches in any dimension	- four points.

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The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth; width less than specified; poor dye penetration, mottled, streaky or cloudy; not singed; excessive neppiness.

5.4.1.2 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified, or more than two yards less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length or if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

5.4.1.3 Shade and appearance examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade and appearance. Any roll in the sample off shade, shaded side to side, side to center, or end to end, shall be cause for rejection of the entire lot represented by the sample.

5.4.1.4 Roll identification examination. During the yard-to-yard examination, each roll in the sample shall be examined for the defects listed below. The lot shall be unacceptable if two or more rolls in the sample contain one or more of the following defects:

The preshrinking process is identified by name or trademark on the cloth, ticket, or package.
Not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.
Face marking missing from either or both ends.

6. PACKAGING

6.1 Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order.

7.2 Approval of quarpel water-repellent and durable flame-resistant treatment formulas. Approval of the such components and combinations and formulas is the responsibility of the U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014. Only those treatments approved and so listed in the invitation for bids or request for proposals shall be considered acceptable for the related procurement.

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7.3 Sources of Government documents. Copies of military and Federal documents are available from:

Department of the Navy
Defense Printing Service
Detachment Office
700 Robbins Avenue, BLDG 4D
Philadelphia, PA 19111-5093

Rules And Regulations Under The Textile Fiber Products Identification Act - Application for copies may be addressed to:

Federal Trade Commission
Washington, DC 20580

7.3.1 Sources of non-Government documents

ASTM Test Methods - Applications for copies should be addressed to:

American Society For Testing and Materials
1916 Race Street
Philadelphia, PA 19103-1187

AATCC Test Methods - Applications for copies should be addressed to:

American Association of Textile Chemists and Colorists
P.O. Box 12215
Triangle Park, NC 27709-2215

Sears Fabric Defect Scales and Fabric Defect Replica Kits are available from:

Sears, Roebuck and Company
Department 817, ATTN: BSC 23-29
Sears Tower
Chicago, IL 60684

MILITARY INTERESTS:

Custodians
Army - GL
Navy - SA
Air Force - 99

Review Activities
Army - MD
Air Force - 45, 82

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
DLA - CT

Project
8305-0529

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